

IRIS: Internal Representation and Insight System

Sequence-Level Evidence for the Linear Representation Hypothesis and a Geometric Instrument for Alignment Monitoring

Tobias Stevenson

Sakin.AI — Safina Ecosystem

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Abstract

We present IRIS (Internal Representation and Insight System), a geometric instrument for observing and monitoring the internal state of language models. IRIS provides initial sequence-level empirical evidence for the linear representation hypothesis — demonstrating that the relational arrangement of tokens in activation space reflects semantic structure without supervision, across model families and processing depths. IRIS extracts six properties from the residual stream of a transformer model at a chosen layer, producing a per-sequence activation constellation — a PCA projection of token representations that reflects the relational geometry of meaning as the model has internally organised it. We demonstrate that this constellation is empirically deterministic under fixed seed conditions (zero variance across repeated extractions), validating IRIS as a reliable measurement instrument.

We report seven empirical findings. In diffusion model space, we demonstrate that visual output similarity does not predict latent geometric proximity — the core empirical argument for monitoring internal geometry rather than output. In language model space, we show that tokenisation boundaries determine which concepts exist as geometric units, that transformer processing proceeds through two distinct geometric regimes (semantic organisation followed by output preparation), and that instruction tuning encodes geometric valence separation invisible to output-level evaluation. Specifically, base TinyStories-33M shows near-zero geometric distance between positive and negative conceptual space (0.000177), while the instruction-tuned variant shows a $\times 997$ expansion (0.176622) — replicated across three independent runs. A seventh finding emerges from systematic multi-model evaluation across 12 models and 252 extractions: within the evaluated Pythia family, scale is associated with amplified geometric divergence following data curation. At 1.4B parameters, deduplication alone produces cross-model constellation distances approaching the theoretical maximum (1.791), compared to negligible divergence (0.100) at 70M parameters — despite zero internal valence separation in either pair.

We argue that output-level alignment monitoring is fundamentally insufficient and that geometric monitoring of internal representations — as implemented in IRIS — provides a complementary and empirically grounded alternative. IRIS is fully open-source and available at github.com/h4dopel0gic/IRIS.

1. Introduction

The linear representation hypothesis — that neural networks encode features as directions in activation space and that semantic structure is linearly preserved in those directions — has accumulated substantial evidence from circuit analysis [4], toy model experiments [16], and activation steering [7]. The existing evidence operates primarily at the level of individual features and circuits: specific directions encode specific concepts, and those directions can be identified, manipulated, and studied.

What remains less explored is whether this structure extends to the sequence level — whether the relational arrangement of all tokens in a sequence, taken together, preserves semantic relationships geometrically. If the linear representation hypothesis holds at sequence level, then the PCA projection of a model’s residual stream across a prompt should show semantically related tokens in geometrically proximate positions, without any supervision signal pointing toward that arrangement. IRIS is designed to test this claim and to use the resulting geometric structure as an alignment monitoring instrument.

The IRIS methodology emerged from empirical work in diffusion model latent space using EAR [Appendix A.], where PCA served as an effective dimensionality reduction convenience for mapping high-dimensional activation structures into observable visual trajectories. While subsequent iterations (IRIS v2) target rotation-invariant metrics like Centered Kernel Alignment (CKA), the current work maintains the PCA-based approach to preserve continuity with the EAR pipeline and to demonstrate the raw, deterministic relational geometry visible in linear projections.

The dominant paradigm for evaluating language model alignment operates at the output level: does the model produce appropriate responses, refuse harmful requests, score well on benchmarks? This paradigm has a fundamental limitation. A model can produce acceptable output while its internal representations have drifted significantly from a trusted reference state. The geometry of meaning within the model — how concepts relate to each other in activation space, which regions are emphasised, which have been suppressed — is not observable from the surface of its outputs.

This paper introduces IRIS (Internal Representation and Insight System), an instrument for observing and monitoring the internal geometric state of language models. IRIS extracts activation constellations — PCA projections of residual stream representations that reveal the relational structure of meaning as a model has organised it internally. The constellation is not a visualisation tool. It is a measurement instrument: deterministic, comparable across models and model versions, and sensitive to structural changes that output evaluation may miss.

Evaluation breadth in this study was intentionally focused on a systematic multi-model battery spanning 12 variants up to 2.7B parameters. These extractions were conducted on consumer-grade hardware (16GB VRAM), demonstrating that significant geometric monitoring and alignment insights are accessible without frontier-scale compute resources. We acknowledge that while these findings are robust at the evaluated scales, further investigation is required to map the geometric thresholds of models exceeding 7B parameters.

The motivation for IRIS arose from empirical work in diffusion model latent space, where we demonstrated that models producing visually similar output can occupy geometrically distant internal regions. This finding — that visual similarity does not predict latent proximity — establishes the core alignment monitoring argument: if two models look the same from the outside but are geometrically different on the inside, any evaluation methodology that only examines outputs will fail to detect the difference. IRIS was designed to make that difference visible and measurable.

We make the following contributions:

- Sequence-level empirical evidence for the linear representation hypothesis: the relational geometry of token activation constellations reflects semantic structure without supervision, across model families, scales, and processing depths.
- A six-property geometric extraction methodology for transformer residual streams, producing per-sequence activation constellations comparable across models and layers.
- Empirical demonstration that the extraction is deterministic under fixed-seed conditions — any measured geometric distance is signal, not instrument noise.
- Evidence that tokenisation boundaries are architectural determinants of conceptual geometry, with implications for cross-model alignment monitoring.
- Evidence that instruction tuning encodes geometric valence structure invisible to output evaluation, demonstrated by a $\times 997$ expansion of positive-negative geometric distance after fine-tuning (replicated across three independent runs).
- An open-source implementation with a full Gradio interface, supporting extraction, constellation visualisation, multi-model comparison, and alignment comparison experiments.

1.1 Organisation

Section 2 reviews related work. Section 3 presents the IRIS methodology. Section 4 reports the empirical findings. Section 5 discusses implications for alignment monitoring. Section 6 addresses limitations and future work. Section 7 concludes.

2. Related Work

2.1 The Linear Representation Hypothesis

The linear representation hypothesis proposes that neural networks represent features as directions in a high-dimensional activation space, and that semantic relationships are preserved linearly in those directions [1]. Early evidence came from word embedding spaces, where arithmetic over direction vectors reflected semantic relationships [1]. Subsequent work extended this to transformer models, showing that attention heads implement identifiable operations [2] and that individual neurons correspond to interpretable features [4].

Elhage et al. [2] provided a mathematical framework for understanding how transformers use the residual stream as a shared communication channel, with each layer reading from and writing to it additively. This framing is foundational for IRIS: the residual stream at any layer is a linear superposition of all preceding computations, and its geometry at any point reflects the model's current internal organisation of the input.

The toy models of superposition work [16] demonstrated that transformers encode far more features than they have dimensions by using approximately orthogonal directions — explaining the geometric structure that IRIS's PCA constellation reveals. Our findings provide sequence-level empirical evidence for this framework: the constellation of token representations in a prompt reflects semantic structure that the model has organised without supervision.

IRIS extends the linear representation hypothesis in two directions. Finding 5 identifies tokenisation as an architectural precondition for concept-level geometric encoding — a concept fragmented by the tokeniser cannot develop a coherent geometric representation regardless of training. Finding 7 shows that scale amplifies the sensitivity of representational geometry to training data distribution, suggesting the geometric structure predicted by the hypothesis becomes more expressive and more volatile at larger parameter counts.

2.2 Mechanistic Interpretability

Mechanistic interpretability seeks to understand how specific computations are implemented within neural networks. Olah et al. [4] demonstrated that individual neurons in vision models correspond to interpretable features, and subsequent work extended this to language models through the identification of circuits — subgraphs of the network responsible for specific capabilities [4]. Nanda et al. [3] provided empirical evidence for the superposition hypothesis, showing that transformer models represent far more features than they have dimensions by exploiting approximately orthogonal directions in activation space.

IRIS complements this tradition but operates at a different level of granularity. Where mechanistic interpretability seeks to identify specific features and circuits, IRIS monitors the relational geometry of the full residual stream across a sequence — the constellation of how all tokens are arranged relative to each other at a given processing depth. This is a coarser instrument by design, suited to monitoring rather than circuit-level analysis.

2.3 Probing and Representation Analysis

Linear probing methods [5] examine whether specific information is linearly decodable from model representations. Representation similarity analysis (RSA) [6] compares representation geometry across models and conditions using correlation of distance matrices. These approaches establish that model representations encode structured semantic information and that this structure is comparable across different systems.

The activation constellation produced by IRIS is related to representation similarity analysis but differs in its application: rather than comparing the geometric structure of representations across tasks or stimuli, IRIS monitors the geometric structure of representations within a sequence across model versions. The focus is temporal (does the geometry change between versions?) rather than structural (what information is encoded?).

2.4 Alignment Evaluation

Current alignment evaluation methods include red-teaming [13], scalable oversight [18], and constitutional AI [14]. These methods operate primarily at the output level, evaluating model behaviour through its responses. Several works have noted limitations of output-level evaluation, including the difficulty of evaluating rare failure modes [15] and the potential for models to behave differently during evaluation than deployment.

IRIS provides a complementary approach: monitoring internal geometric state rather than evaluating outputs. The most closely related work is activation steering [7], which modifies model behaviour by adding vectors to the residual stream during inference. Where activation steering intervenes in the residual stream, IRIS observes it. Both approaches demonstrate that the residual stream is a meaningful locus of model behaviour.

2.5 Diffusion Model Latent Space Analysis

Several works have examined the structure of diffusion model latent spaces [9] and demonstrated that semantic information is encoded in intermediate representations. The finding that CLIP embeddings align text and image representations in a shared semantic space [8] is related to the IRIS argument: if semantic similarity is reflected in geometric proximity in one embedded space, divergence in that space reflects semantic divergence.

Our empirical foundation in diffusion space (Section 4.1-4.3) extends this line of work by demonstrating that training lineage is reflected in latent geometry and that visual output similarity is not predictive of latent proximity — a result that motivates the IRIS approach to language model monitoring.

3. Methodology

IRIS extracts six geometric properties from the residual stream of a transformer language model at a specified layer. Together these properties constitute the activation constellation — the geometric signature of how the model has organised a sequence of tokens internally at that processing depth.

3.1 Extraction Properties

Let $R \in \mathbb{R}^{\text{seq} \times d_{\text{model}}}$ denote the residual stream at layer l for a sequence of seq tokens, where d_{model} is the model's hidden dimension. IRIS extracts the following six properties. Key metrics are defined formally below.

- Cosine distance between two vectors a and b : $d_{\text{cos}}(a,b) = 1 - (a \cdot b) / (||a|| \cdot ||b||)$
- Sequence centroid: $\mu = (1/\text{seq}) \sum_i r_i$ where r_i is the residual vector for token i
- Valence Bias Index: $VBI(v_1,v_2) = d_{\text{cos}}(\mu_{\{v_1\}}, \mu_{\{v_2\}})$ where μ_v is the mean centroid across all prompts in valence class v
- Noise floor: $\sigma^2_{\text{instrument}} \approx 0$ under fixed seed conditions (verified empirically in Finding 4)

IRIS extracts the following six properties:

Property	Definition	What It Discloses
RMS Magnitude	$ r_i _2$ for each token i	Representational energy / salience
Layer Delta Magnitude	$ r_i^l - r_i^{l-1} _2$	Where meaning crystallises across depth
Attention Entropy	$H(A_i) = -\sum_j a_{ij} \log a_{ij}$	Confidence and focus of attention
Sequence Centroid	$\mu = (1/\text{seq}) \sum_i r_i$	Absolute position of constellation
Cosine Distance	$1 - \cos(r_i, \mu)$	Token deviation from sequence centre
PCA Projection	$\text{PCA}(R, n=2) \rightarrow [\text{pc1}_i, \text{pc2}_i]$	Relational geometry — the constellation

Table 1. Six geometric properties extracted by IRIS per token per layer.

IRIS does not claim direct semantic equivalence between geometric proximity and human conceptual meaning. Rather, IRIS measures representational structure correlated with token processing within the model. Proximity in the constellation reflects co-variation in activation space, not verified conceptual understanding. This distinction is maintained throughout.

3.2 The Activation Constellation

The PCA projection is the most informationally significant property. While other properties are scalar and per-token, the PCA projection is relational — it reveals the shape of the sequence, how tokens are arranged with respect to each other in the model's internal space. This is not a property of any individual token but of the whole sequence.

Formally, let $C = \text{PCA}(R, n=2) \in \mathbb{R}^{\{\text{seq} \times 2\}}$ denote the constellation. The top principal component PC1 captures the axis of maximum variance across tokens; PC2 captures the orthogonal axis of next-greatest variance. Tokens that are geometrically proximate in activation space remain proximate in the projection; tokens the model has differentiated are separated.

For cross-model comparison, constellations are normalised to zero mean and unit variance per axis before comparison. This removes differences in scale and position, enabling shape comparison that is independent of the absolute coordinate system.

3.3 Attention Entropy as CFG Driver

This section describes a principled mapping from IRIS activation properties to diffusion model conditioning parameters, used in the EAR pipeline (Appendix A, github.com/h4dopel0gic/EAR). EAR is the diffusion rendering system that preceded and motivated IRIS — the empirical foundation for the alignment monitoring argument in Findings 1–3 was established there.

The mapping from attention entropy to classifier-free guidance (CFG) scale is principled rather than heuristic. Low attention entropy reflects focused, confident attention — the model is attending sharply to specific tokens. This corresponds to a strong guidance signal: high CFG. High attention entropy reflects diffuse, uncertain attention, corresponding to low guidance. This mapping is used in the IRIS-to-EAR bridge (Appendix A).

3.4 Determinism and the Noise Floor

Accurate alignment monitoring requires that the measurement instrument itself contributes no variance. A non-deterministic extractor would conflate instrument noise with genuine model drift.

We establish the IRIS noise floor through five repeated extractions on the same model (GPT-2), prompt, layer, and random seed. All sources of randomness are fixed: Python's random module, NumPy, and PyTorch seeds are set identically before each extraction. PCA is fit on the same data in the same order.

Results are reported in Finding 4. The instrument satisfies $\sigma^2_{\text{instrument}} \approx 0$ under fixed-seed conditions: zero variance at numerical precision across all five runs and all six extracted properties. This establishes that any measured geometric distance between two model states is attributable to genuine representational difference rather than instrument noise.

3.5 Implementation

IRIS is implemented in Python using TransformerLens [12] for model loading and residual stream access. Extraction, visualisation, and comparison are provided as standalone scripts and integrated into a Gradio web interface. The interface supports single-model constellation viewing, multi-model comparison with normalisation toggle, alignment comparison across valence classes, and drift monitoring between constellation files.

Supported models include GPT-2 family, Pythia family, TinyStories family, and any model supported by TransformerLens. Large models (7B+) are loaded in float16 automatically.

4. Empirical Findings

We report seven empirical findings across two experimental domains. Findings 1–3 were produced using a custom pre-VAE latent extraction pipeline within EAR (Embedded-space Audio-induced Reflections), the diffusion system that preceded and motivated IRIS (github.com/h4dopel0gic/EAR). They establish the geometric monitoring argument in a tractable empirical setting. Findings 4–7 were produced using IRIS directly, applied to transformer language models. Findings 4–7 are reported from primary extraction sessions and confirmed across multiple independent batch runs (replication_v01 through replication_v02, 252 total extractions across 12 model variants).

4.1 Finding 1: Visual Similarity Does Not Predict Latent Proximity

This finding and Findings 2–3 were produced in the EAR pipeline using a custom ComfyUI extraction node that captures pre-VAE latent tensors from diffusion model inference runs. These findings motivated the development of IRIS as a dedicated geometric monitoring instrument for language models; they demonstrate the core argument — that internal geometry diverges independently of outputs — in a tractable and visually interpretable domain before extending it to transformer residual streams.

We compare three diffusion models — Stable Diffusion 1.5 (SD1.5) [9], RealisticVision, and PonyDiffusion — using identical inputs: the same seed, prompt ("neutral"), CFG scale, denoise strength, and step count. Pre-VAE latent tensors are saved per model using a custom ComfyUI node. We compute pairwise cosine distance and L2 distance between these tensors.

Model Pair	Cosine Distance	L2 Distance	Interpretation
SD1.5 ↔ RealisticVision	0.5945	2055.1	Moderate divergence — different fine-tune domain

Model Pair	Cosine Distance	L2 Distance	Interpretation
SD1.5 ↔ PonyDiffusion	0.4866	1957.8	Closest pair — shared architectural lineage
RealisticVision ↔ PonyDiff	0.6693	1263.3	Most distant — maximally divergent geometry

Table 2. Pairwise latent distances between diffusion models under identical inputs.

PonyDiffusion and RealisticVision produce visually similar figurative output but occupy the most geometrically distant latent positions (cosine distance 0.6693). SD1.5 and PonyDiffusion are geometrically closest (0.4866) despite very different outputs. Visual similarity is not predictive of latent proximity. This finding motivates the core IRIS argument: output-level monitoring cannot detect representational divergence.

4.2 Finding 2: Internal Geometry Reflects Training Lineage

The geometric proximity of SD1.5 and PonyDiffusion reflects their shared training lineage: PonyDiffusion is fine-tuned from an SD1.5 base. Despite dramatically different visual output, PonyDiffusion retains geometric proximity to its base model. RealisticVision, fine-tuned on a more specialised data distribution, has drifted further from the SD1.5 origin.

This finding establishes that latent geometry encodes training history. A baseline constellation extracted at a known-good checkpoint provides a reference against which subsequent versions can be compared. Drift from this reference is a measurable geometric quantity.

4.3 Finding 3: Representational Bias Concentrates in Specific Subspaces

Per-channel cosine distance analysis reveals that geometric divergence is not uniformly distributed across latent channels. Channel 1 (associated with compositional structure) reaches cosine distance 0.8039 between RealisticVision and PonyDiffusion — near-orthogonal, indicating maximal divergence along this axis. Other channels show more moderate divergence.

This finding has practical implications for alignment monitoring: probes and anchors can be targeted at the most sensitive subspaces rather than applied uniformly. Channel-specific monitoring is more efficient than global monitoring.

From Diffusion to Language Models

Findings 1–3 establish the core geometric monitoring argument in diffusion model latent space: internal representations diverge independently of visual output, training history is encoded in geometry, and divergence concentrates in specific subspaces. This is a tractable empirical domain — diffusion latents are high-dimensional but the output is directly observable as an image, making the relationship between internal geometry and surface behaviour visually interpretable.

The same argument applies with greater urgency to language models, where the gap between internal geometry and output behaviour is harder to observe directly. A language model producing harmful or misaligned behaviour may present acceptable outputs during evaluation. A monitoring instrument that reads internal geometry rather than evaluating outputs addresses this gap directly.

IRIS was built to pursue this argument in transformer residual streams. Before reporting substantive findings, Finding 4 establishes the instrument’s reliability: any measured geometric distance must be attributable to genuine representational difference rather than instrument noise. This is the prerequisite for all subsequent measurements.

4.4 Finding 4: The Instrument is Deterministic

Five repeated extractions of the IRIS constellation from GPT-2 (layer 6) on the Basmala prompt ("In the name of God, the most gracious") under fixed seed (42) yield the following results:

Metric	Value	Interpretation
Centroid norm std	0.000000	Zero variance across all five runs
Pairwise cosine distance mean	0.00000000	Perfect identity between runs
Pairwise cosine distance max	0.00000000	No deviation at any precision
Per-token PC1 std	0.000000	All tokens: zero coordinate variance
Per-token PC2 std	0.000000	All tokens: zero coordinate variance
Per-token RMS std	0.000000	All tokens: zero magnitude variance

Table 3. Noise floor results from five repeated extractions under fixed seed.

The instrument is not merely stable — it is deterministic. Any measured geometric distance between two model versions is pure signal. The instrument contributes zero variance to the measurement.

4.5 Finding 5: Tokenisation Shapes Conceptual Geometry

We compare the activation constellations of GPT-2 and Mistral-7B [10] on the same Basmala prompt at equivalent proportional layer depths (GPT-2 layer 6 of 12; Mistral-7B layer 16 of 32). After normalising both constellations to zero mean and unit variance, we overlay them and compare token positions.

In GPT-2, which tokenises 'gracious' as a single token, 'gracious' and 'God' are geometrically proximate — a representational proximity emerges between these tokens without supervision. In Mistral-7B, which tokenises 'gracious' as 'gr' + 'acious', this pairing is disrupted. The subword fragments land in different regions of the constellation, separated from 'God'. The projected representation associated with the tokenised form of 'gracious' no longer occupies a region proximate to 'God' in the activation geometry.

This finding has a direct implication for alignment monitoring: concept-level geometric anchoring requires either tokeniser-invariant representations or explicit handling of subword fragmentation. A drift monitor anchored to 'gracious' as a geometric unit cannot directly compare models with different tokenisation schemes.

Tokenisation is not neutral. It is an architectural decision that determines which concepts exist as geometric units and which are fragmented. A model's internal organisation of meaning is downstream of its tokeniser.

4.6 Finding 6: Instruction Tuning Correlates With Geometric Valence Separation

We compare TinyStories-33M (base) [11] against TinyStories-Instruct-33M (instruction-tuned) across three prompt valence classes: positive, neutral, and negative (5 prompts each, 15 total per model). Both models share identical architecture and tokeniser; the only difference is instruction fine-tuning.

For each model, we compute the mean constellation centroid per valence class and measure pairwise cosine distances between class centroids — the Valence Bias Index. A model that processes positive and negative content with similar internal geometry has low positive-negative distance. A model that has developed internal valence structure has high distance.

Valence Pair	Base Model	Tuned Model	Change
Positive ↔ Negative	0.000177	0.176622	×997
Positive ↔ Neutral	0.660846	0.046977	−93%
Negative ↔ Neutral	0.643203	0.043409	−93%

Table 4. Valence Bias Index: within-model cosine distance between valence class centroids. Values from replication_v02 batch run. Finding replicates across three independent extraction runs (×1,156, ×997, ×997).

The base model treats positive and negative prompts with near-identical internal geometry (Pos↔Neg: 0.000177) and sits roughly equidistant from positive and negative space — symmetric and geometrically neutral. The instruction-tuned model has developed a $\times 997$ expansion of positive-negative geometric distance (0.176622), while simultaneously collapsing its Pos↔Neutral and Neg↔Neutral distances by approximately 93%. The tuned model departs substantially from the comparatively symmetric geometry of the base model: its neutral representational space has been pulled inward toward both valence poles, indicating a broad reorganisation of valence-associated geometry that extends beyond output behaviour.

This finding is significant for alignment monitoring because it demonstrates that instruction tuning produces structural changes in internal representation that are:

- Invisible to output evaluation (both models produce coherent, appropriate text)
- Detectable by IRIS without examining any output token
- Quantifiable as a specific geometric distance metric
- Interpretable in terms of the model's internal organisation of conceptual space

The greatest cross-model geometric divergence occurs on neutral prompts (mean cosine distance 1.382, vs 1.285 for positive and 1.245 for negative prompts from replication_v02). The instruction-tuned model's neutral representational space has been reorganised most dramatically.

Output Similarity vs Internal Divergence

This finding operationalises the paper's central thesis. Both TinyStories-33M and TinyStories-Instruct-33M produce coherent children's stories. On the 15 experimental prompts, both produce qualitatively similar output in register, vocabulary, and narrative structure. An output-level evaluator sees two comparable story-generating models.

The cross-model geometric distances on these identical prompts are mean 1.285 on positive prompts, 1.382 on neutral, and 1.245 on negative — on a scale where 2.0 is maximum cosine distance. The within-model valence separation of the base model is 0.000177. The cross-model geometric distance is approximately $7,800\times$ larger. Models producing qualitatively similar output occupy dramatically different activation space regions. Output similarity does not imply representational similarity. IRIS detects the difference. Output evaluation may not.

4.7 Finding 7: Scale Amplifies Training Data Curation Effects

We compare Pythia base models against their deduplication-trained counterparts across three scale points, using the same 15-prompt valence battery. Deduplication removes near-duplicate documents from training data but does not introduce any alignment-directed training signal.

Model Pair	VBI Pos↔Neg	Cross-Model Dist (Neutral)	Interpretation
Pythia-70M vs Deduped	0.000	0.100	Negligible geometric difference
Pythia-160M vs Deduped	0.000	0.540	Moderate geometric divergence
Pythia-1.4B vs Deduped	0.000	1.791	Near-maximum geometric divergence

Table 5. Pythia base vs deduped Valence Bias Index and cross-model distances across three scale points.

All three Pythia pairs show zero Valence Bias Index — correctly. Deduplication is not an alignment-directed intervention and produces no internal valence geometry. However, cross-model geometric distance increases dramatically with scale: from 0.100 at 70M parameters to 1.791 at 1.4B — approaching the theoretical maximum cosine distance of 2.0 on neutral prompts.

This finding has two implications. First, IRIS correctly distinguishes between training interventions that do and do not produce valence geometry — validating the instrument's sensitivity and specificity. Second, scale amplifies the geometric effect of training data curation: the same data cleaning intervention that leaves small models geometrically nearly identical produces fundamental representational divergence at larger scale.

Within the Pythia model family, scale is associated with amplified geometric divergence following deduplication. What produces negligible geometric change at 70M parameters produces near-maximum constellation divergence at 1.4B. IRIS detects this. Output evaluation would not.

Practical implication: geometric drift thresholds appropriate for small models may be far too conservative for large models. Alignment monitoring frameworks must account for scale-dependent sensitivity.

4.8 Observation: Peak Delta and Training Data Quality

During systematic multi-model evaluation, peak layer delta magnitude was observed to increase with model scale across the Pythia family — consistent with dimensional scaling (larger residual streams produce larger absolute delta norms). However, Phi-2 (2.7B parameters, trained on synthetic educational content) showed peak delta comparable to GPT-2-Large (774M parameters, trained on raw web text) — approximately equal or lower despite being 3.5× larger by parameter count.

We hypothesise that peak delta reflects training data structure as well as model scale. Models trained on clean, structured data (Phi-2: textbooks and educational content curated by Microsoft) require less per-layer geometric work to organise representations than models trained on noisy web data.

The residual stream does not need to travel as far per layer to produce coherent semantic representations because the learned representations are better conditioned from training.

This observation is preliminary and requires controlled comparison across matched model sizes with different training data distributions. If confirmed, peak delta would constitute a diagnostic signal for training data quality — accessible without training data access, from the model's geometric processing signature alone.

5. Discussion

5.1 IRIS and the Linear Representation Hypothesis

The linear representation hypothesis predicts that semantic structure is preserved in activation space geometry. IRIS provides the first systematic sequence-level test of this prediction. The activation constellation — the PCA projection of all token representations in a prompt — reveals that semantically related tokens occupy geometrically proximate positions without any supervision signal directing them there. In the reference extraction (Basmala prompt, GPT-2 layer 6), ‘God’ and ‘gracious’ cluster together in the constellation’s lower region. The representations associated with mercy-related and divinity-related tokens occupy proximate regions in the constellation without supervision. This is consistent with the linear representation hypothesis at sequence level.

Finding 5 extends the hypothesis by identifying tokenisation as an architectural precondition for the emergence of token-level representational coherence. The hypothesis describes how concepts are represented as directions. Finding 5 asks: what determines which concepts can develop directions? The answer is the tokeniser. ‘Gracious’ as a unit develops a direction that neighbours ‘God’. ‘Gr’ and ‘acious’ as fragments cannot. This is not a limitation of the model’s training but an architectural constraint upstream of it.

Finding 7 extends the hypothesis by showing that the geometric expressiveness it predicts scales with model size in a non-trivial way. Data curation effects that are geometrically invisible at 70M parameters become near-maximal at 1.4B. The geometric structure the hypothesis predicts is not equally present at all scales — it becomes richer and more sensitive as capacity increases.

5.2 The Case for Geometric Monitoring

The seven findings together constitute a coherent empirical argument. Findings 1-3 establish, in a tractable empirical setting (diffusion models), that internal geometry diverges independently of output behaviour, that training history is encoded in geometry, and that divergence is not uniform but concentrated in specific subspaces. Finding 4 establishes that the measurement instrument is reliable. Findings 5-7 extend the methodology to language models and demonstrate that

tokenisation, fine-tuning, and training data curation all produce geometric changes invisible to output evaluation.

The argument is not that output evaluation is wrong. It is that output evaluation is incomplete. A model that produces acceptable output can simultaneously have internal representations that have drifted from a reference state. Geometric monitoring provides a complementary signal that is sensitive to changes output evaluation misses.

Representational similarity cannot be reliably inferred from output similarity alone. Output resemblance is an insufficient proxy for representational proximity.

5.3 The Geometric Anchor

Finding 6 suggests a practical monitoring methodology: extract the activation constellation of a model at a known-good state, store it as a geometric reference, and compare subsequent versions against this reference. A drift threshold can be established from the noise floor result — any distance above numerical precision represents genuine model change.

The Valence Bias Index introduced here is one operationalisation of this approach. A more comprehensive implementation would monitor the full constellation shape across a battery of anchor prompts, computing rotation-corrected distance to the reference after aligning coordinate systems via a shared anchor.

5.4 Tokenisation as an Alignment Concern

Finding 5 identifies tokenisation as an underappreciated architectural determinant of alignment. The finding that mercy no longer geometrically neighbours divinity in a model that fragments 'gracious' is not a trivial observation about linguistic encoding. It suggests that models with different tokenisation schemes may internally organise semantically related concepts differently — not because of differences in training values, but because of differences in the atomic units from which internal representations are built.

This has implications for any alignment monitoring methodology that relies on tracking specific concepts geometrically. Cross-model comparison requires either models sharing a tokeniser or explicit handling of concept fragmentation. We flag this as an open problem requiring further investigation.

5.5 Processing Depth and Regime Transitions

The layer sweep comparison (GPT-2 layer 6 vs layer 11) reveals that the activation constellation is not static across processing depth. At layer 6 (mid-network), the constellation shows semantic organisation: one dominant PCA axis (99.7% variance), concept clustering ('God' and 'gracious' proximate), and processing concentrated on semantic pivots. At layer 11 (final layer), the geometry expands (PC1 drops to 40.1%, PC2 rises to 24.1%), the predictive anchor token ('name') is elevated, and function words collapse into a tight cluster.

This suggests that different layers are appropriate for different monitoring tasks. Mid-network layers are more informative about semantic organisation; late layers are more informative about task-directed processing. A comprehensive monitoring methodology may need to monitor at multiple depths.

6. Limitations and Future Work

6.1 PCA as Projection Convenience

PCA is used in IRIS as a dimensionality reduction convenience, not as a privileged semantic basis. The principal components are mathematically valid but geometrically arbitrary: the same underlying activation structure can be represented at any rotation. This introduces two limitations that should be explicitly acknowledged.

First, rotational instability: PCA axes can rotate across different runs on different models, introducing coordinate system ambiguity in cross-model comparisons. Normalisation (zero mean, unit variance) addresses scale and position but not rotation. The Valence Bias Index is computed from centroid distances and is rotation-invariant; the cross-model cosine distances reported may partially reflect axis rotation rather than genuine representational divergence.

Second, linearity: PCA is a linear projection. Activation space is not strictly linear. Non-linear dimensionality reduction methods — UMAP, t-SNE, and manifold methods — preserve local structure that PCA may distort. The current findings are based on linear projections and should be interpreted accordingly.

Several methodological extensions would address these limitations in future work. Procrustes alignment minimises the rotational difference between two constellations by finding the optimal orthogonal transformation — this would enable rotation-corrected cross-model comparison. Centred Kernel Alignment (CKA) [6] is rotation-invariant by construction and provides a principled similarity measure between representation spaces. Singular Vector Canonical Correlation Analysis (SVCCA) captures alignment between subspaces rather than point clouds. Earth Mover Distance (Wasserstein distance) measures distributional rather than centroid-level divergence. Any of these would provide a more rigorous geometric comparison than the current normalised PCA approach, which arose from the preceding EAR [Appendix A.] diffusion parameters. We flag these as the primary methodological development targets for IRIS v2.

6.2 PCA Rotational Instability — Semantic Compass Proposal

PCA axes are mathematically valid but geometrically arbitrary: the same underlying activation structure can be represented at any rotation. This introduces ambiguity in cross-model constellation

comparisons — geometric divergence between models may partly reflect axis rotation rather than genuine representational difference. Normalisation (zero mean, unit variance) addresses scale and position but not rotation.

A proposed resolution is the semantic compass: a carefully selected anchor prompt (the Basmala is a natural candidate given existing extractions) establishes a reference constellation for each model. The rotation matrix between base and comparison model constellations on this anchor is computed and applied before comparison, orienting both models to the same geometric coordinate system. This is left for future work.

6.3 Absolute Distance Values

The absolute cosine distances reported in Finding 6 (1.17-1.34) are large relative to typical representation similarity metrics. This reflects that TinyStories models, being small and domain-specific, occupy tightly constrained PCA spaces that do not align well across models even after normalisation. The scientific interpretation relies on relative patterns across valence classes and the Valence Bias Index, not on absolute distance magnitudes. Future work should investigate normalisation schemes that produce comparable absolute values across model families.

6.4 Model and Prompt Scale

The primary alignment finding (Finding 6) uses small models (TinyStories at 33M parameters). Finding 7 extends evaluation to 1.4B parameters via the Pythia family, demonstrating that the methodology scales and produces meaningful results at larger scale. Systematic evaluation across 12 models from 33M to 2.7B parameters (Phi-2) confirms that IRIS extraction operates correctly across model families and scales. The alignment comparison (Finding 6) has not yet been replicated with models above 33M parameters — this is the primary remaining scale question. We note that Mistral-7B showed similar PC1 dominance to GPT-2 at equivalent proportional depth, suggesting core geometric properties are broadly robust across scale.

6.5 Attention Entropy NaN Values

For TinyStories models, attention entropy values are NaN due to architectural differences in attention pattern caching between model families. The core geometric findings (constellation shape, cosine distances, Valence Bias Index) are computed from the PCA projection and are unaffected. Architecture-aware cache key detection is planned for the next version of the extractor.

6.6 Failure Cases and Boundary Conditions

A credible instrument must demonstrate not only successful measurement but also the conditions under which it fails or produces uninformative results. IRIS has four identifiable failure modes worth documenting explicitly.

Randomised Token Sequences

IRIS was applied to five prompts constructed from randomly shuffled tokens from the existing prompt battery (e.g. 'the sea lost was hope lost the'). At GPT-2 layer 6, the resulting constellations show no interpretable clustering: PC1 explained variance drops from 99.7% on the Basmala to 41–48% on random sequences, and centroid distances between valence classes are near-zero. This is the expected behaviour: IRIS produces uninformative geometry when the input has no semantic structure. The instrument is not finding patterns everywhere indiscriminately.

PCA Collapse Under Semantic Uniformity

Prompts consisting entirely of tokens from a single semantic domain — e.g. the semantic probe 'God mercy compassion grace love' — produce constellations with high PC1 variance (all tokens cluster tightly along one axis) and near-zero PC2 variance. The constellation collapses to approximately one dimension. This is geometrically correct — semantically uniform inputs have low intrinsic dimensionality — but means that two-dimensional PCA projections become uninformative for such inputs. Cross-model comparison on semantically uniform prompts should be interpreted with caution.

Cross-Tokeniser Comparison Breakdown

Finding 5 documents this failure mode directly. When comparing models with different tokenisers on the same text, semantic concepts fragmented differently by each tokeniser cannot be compared at the concept level. The token 'gracious' in GPT-2 and the fragment pair 'gr' + 'acious' in Mistral-7B produce constellations that cannot be meaningfully aligned at the token level. Cross-model comparison is valid when models share a tokeniser; it requires careful handling when they do not.

Late-Layer Instability

At the final one or two layers of any evaluated model, the layer delta profile shows high variance and the PCA constellation becomes less stable across prompt variations. The model is preparing outputs rather than organising representations, and the geometry reflects task-specific rather than semantic-structural organisation. IRIS is most informative at mid-network depths (50–60% of total layers). Extraction at the final layer is not recommended for semantic monitoring purposes.

6.7 Future Work

- Controlled output similarity experiment: paired generation from base and fine-tuned models on held-out prompts, with explicit BLEU/semantic similarity scoring alongside constellation divergence measurement. This would provide the most direct operationalisation of the output monitoring insufficiency claim.
- Non-linear dimensionality reduction: UMAP and t-SNE comparison with PCA constellations to identify structure that linear projection misses.
- Procrustes-corrected cross-model comparison: rotation-corrected constellation alignment using shared anchor prompts as reference orientation.
- Constellation trajectory analysis: monitoring the layer-by-layer geometric arc of a sequence rather than a single-layer snapshot, enabling observation of how processing regimes evolve.
- Semantic compass implementation: rotation-corrected cross-model comparison via shared anchor prompts.

- Larger model evaluation: applying IRIS to GPT-2-XL, Pythia-6.9B, and Mistral-7B at full depth.
 - Multilingual analysis: comparing activation constellations for the same semantic content in different languages and scripts.
 - Bridge 4 translator: a trained mapping from IRIS activation geometry to diffusion conditioning vectors, enabling pure geometry-driven rendering without architectural vocabulary.
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7. Conclusion

We have presented IRIS, a geometric instrument for observing and monitoring the internal state of language models. IRIS provides sequence-level empirical evidence for the linear representation hypothesis and demonstrates its applicability to alignment monitoring. Through seven empirical findings spanning diffusion and language model domains, and systematic evaluation across 12 models and 252 extractions, we have demonstrated that:

- Internal geometry diverges independently of output behaviour — visual or textual similarity is not a reliable proxy for representational similarity.
- The activation constellation provides sequence-level empirical evidence for the linear representation hypothesis: semantically related tokens occupy geometrically proximate positions without supervision, across model families and scales.
- The IRIS extraction is empirically deterministic under controlled extraction conditions, validating its use as a measurement instrument.
- Tokenisation is an architectural determinant of conceptual geometry with implications for cross-model alignment monitoring.
- Instruction tuning encodes geometric valence structure that is invisible to output evaluation, detectable by IRIS through the Valence Bias Index — replicated across three independent extraction runs.
- Scale amplifies the geometric effects of training data curation: deduplication produces negligible geometric change at 70M parameters but near-maximum constellation divergence at 1.4B parameters.

These findings support the argument that geometric monitoring of internal representations is a necessary complement to output-level alignment evaluation. IRIS provides a practical, open-source implementation of this approach, validated empirically and ready for extension to larger models, longer sequences, and continuous monitoring applications.

The instrument is publicly available at github.com/h4dopel0gic/IRIS. We invite the interpretability and alignment communities to build on this work, challenge these findings, and extend the methodology.

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Appendix A: EAR Pipeline and Bridge Architecture

IRIS grew from EAR (Embedded-space Audio-induced Reflections), an audio-guided diffusion pipeline that provided the empirical foundation for the alignment monitoring argument. EAR demonstrates that activation geometry can be measured, compared, and monitored in a tractable empirical setting. The findings from EAR Experiment 4 (Findings 1-3 in this paper) motivate the extension to language models.

The IRIS timeline format is EAR-compatible: IRIS activation extractions produce timeline JSON files that the EAR render pipeline can consume directly, enabling visualisation of language model activation geometry through diffusion rendering. This bridge is used for demonstration purposes; the core IRIS findings are independent of EAR.

EAR is publicly available at github.com/h4dopel0gic/EAR. The repository includes the full audio-guided diffusion pipeline, ComfyUI workflow, custom pre-VAE latent extraction node, and the EAR_Echodrop LoRA. Findings 1–3 in this paper are fully reproducible using the EAR codebase.

Appendix B: Prompt Battery for Finding 6

Positive Valence

(1) This is a wonderful day full of hope and possibility. (2) The results exceeded all expectations and everyone celebrated. (3) She felt grateful and joyful walking through the garden. (4) The team succeeded brilliantly and the future looks bright. (5) Love and kindness filled the room as they reunited.

Neutral Valence

(1) In the name of God, the most gracious. (2) The train arrives at the station every morning. (3) Water flows downhill toward the sea. (4) The document was placed on the table. (5) She walked across the room and opened the window.

Negative Valence

(1) The situation deteriorated rapidly and all hope was lost. (2) Failure and disappointment defined the entire experience. (3) The damage was severe and recovery seemed impossible. (4) Fear and despair spread through the crowd. (5) The loss was devastating and the pain overwhelming.